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Studierichting: Informatica

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$$\begin{aligned}& \sum_{n=1}^{\infty} \frac{3^{2n}}{n!} \\& \lim_{n \rightarrow +\infty} \left| \frac{\frac{3^{2(n+1)}}{(n+1)!}}{\frac{3^{2n}}{n!}} \right| \\&= \lim_{n \rightarrow +\infty} \left| \frac{3^{2(n+1)}}{(n+1)!} \cdot \frac{n!}{3^{2n}} \right| \\&= \lim_{n \rightarrow +\infty} \left| \frac{3^{2n+2}}{3^{2n}} \cdot \frac{n!}{(n+1)n!} \right| \\&= \lim_{n \rightarrow +\infty} \frac{9}{n+1} \\&= 0\end{aligned}$$

Omdat $0 < 1$, is de reeks convergent volgens het Criterium van d'Alembert.

References